DBD-CR: Dual-Branch Diffusion Residual Reconstruction for Cloud Removal in Optical Remote Sensing Images

Chen Yang, *Member, IEEE*, Qinxu Gao, Guanhua Chen, and Huiying Li

***Abstract*— Optical remote sensing image cloud removal is a crucial area of research within remote sensing technology. Its primary goal is to recover surface information from images obscured by clouds. Although existing generative models have achieved significant success in cloud-free image reconstruction, providing important support for downstream tasks, challenges remain in capturing and reconstructing complex detailed features. To address these issues, this article presents a dual-branch diffusion residual reconstruction framework for optical remote sensing image cloud removal, called the DBD-CR framework. Specifically, in the initial reconstruction stage, we utilize a diffusion model to obtain the initial reconstructed image from the cloudy input. Next, in the discrepancy correction stage, we calculate the residual texture map between the cloudy image and the initial reconstructed image to capture the details of the surface features that have not been restored. In addition, we integrate a frequency domain enhancement interaction attention module and an adaptive mixed attention module (AMAM) into the denoising diffusion model. These components capture crucial details in the frequency domain and adaptively select features that combine downsampling and upsampling. By adopting more comprehensive evaluation metrics and conducting extensive experiments on three public remote sensing cloud removal datasets, the DBD-CR framework achieves a maximum reduction of 18.081 in the Fréchet inception distance (FID) metric and a maximum improvement of 0.902 in the peak signal-to-noise ratio (PSNR) metric, further demonstrating its advanced performance for optical remote sensing image cloud removal tasks.**

***Index Terms*— Cloud removal, dual-branch diffusion, optical remote sensing images, residual reconstruction.**

I. Introduction

REMOTE sensing technology refers to the comprehensive techniques used to obtain information about the Earth's surface, atmosphere, and oceans through various sensors. As a key component of remote sensing technology, optical remote sensing images leverage optical sensors to capture surface features using visible light and other spectral bands. They hold significant potential for a wide range of applications, including agriculture and forestry [1], [2], urban planning [3], and environmental monitoring [4]. However, optical remote sensing images have consistently been affected by meteorological conditions, such as cloud cover. According to observations from the Moderate Resolution Imaging Spectroradiometer, approximately 55% of the land surface and about 72% of the ocean are covered by clouds [5]. The presence of clouds reduces the quality of optical remote sensing images, significantly hindering the analysis and application of remote sensing data. Therefore, effectively removing clouds from optical remote sensing images is essential to enhance remote sensing data utilization and lays the foundation for improving downstream tasks that involve optical remote sensing images.

In recent years, the emergence of generative adversarial networks (GANs) has provided new insights to address many challenging problems in the field of image processing. GANs can generate high-quality images through adversarial training of generators and discriminators, providing a powerful tool for tasks such as image super-resolution [6], image restoration [7], and style transfer [8]. Therefore, some researchers have begun to explore the application of GANs to the cloud removal task in optical remote sensing images. These methods [9], [10], [11], [12], [13], [14], [15] leverage the generative capabilities and adversarial training mechanisms of GANs to effectively recover some clear information from images obscured by cloud cover. Although the cloud removal method based on GANs for optical remote sensing images can reconstruct cloud-free images for restoring landforms, it is difficult to control the training balance by training two networks simultaneously with GANs. The training is unstable and the model is easy to collapse. In addition, the methods based on GANs, due to the tendency of the generator to fill the region with simple patterns, lead to the generation of images with smooth transitions or with colors that do not match the reconstructed landforms.

With continuous exploration in the field of image generation, diffusion models have gradually emerged as a promising generation method. They transform real images into random noise by gradually adding noise and learn how to reverse denoising to gradually recover a clear image from the noise [16]. Since the diffusion model can effectively improve the shortcomings of GANs methods, many researchers have begun to explore cloud removal methods based on the diffusion model for optical remote sensing images. Denoising diffusion probabilistic model (DDPM)-CR [17] integrates optical and SAR images using a denoising diffusion model and extracts

Received 1 January 2025; revised 25 February 2025; accepted 8 March 2025. Date of publication 11 March 2025; date of current version 10 April 2025. This work was supported by Scientific and Technological Development Scheme of Jilin Province under Grant 20230201089GX. *(Corresponding author: Huiying Li.)*

Chen Yang, Qinxu Gao, and Guanhua Chen are with the College of Computer Science and Technology, Jilin University, Changchun 130012, China.

Huiying Li is with the College of Computer Science and Technology and the Key Laboratory of Symbol Computation and Knowledge Engineer of Ministry of Education, Jilin University, Changchun 130012, China (e-mail: lihuiying@jlu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3550422

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

depth features to recover information from regions obscured by clouds. DiffCR [18] achieves efficient cloud removal from optical satellite images by decoupling the encoder and temporal conditional fusion blocks, combined with conditionally guided diffusion and deep convolutional networks. IDF-CR [19] achieves two-stage processing from initial cloud removal to detail optimization by combining a cloud removal module in pixel space with an iterative noise diffusion network in potential space and an unsupervised iterative noise refinement module. Although methods based on diffusion models show significant advantages in cloud removal, these methods greatly neglect the complex details of surface features in the generated cloud-free images, leading to a degradation of the quality of the generated images.

To address the aforementioned problems, we propose a dual-branch diffusion residual reconstruction framework for cloud removal in optical remote sensing images (DBD-CR). Specifically, the DBD-CR framework begins with the initial reconstruction stage, where a cloudy image is input into the diffusion model to generate an initial reconstructed image. Although the initially generated cloud-free image roughly restores the original scene, it may lack certain details. Next, in the discrepancy correction stage, we calculate the residuals between the original cloudy image and the initial reconstructed image, inputting these results into the diffusion model to obtain a residual texture map. This residual texture map primarily contains the details of surface features that have not been restored. The DBD-CR framework then fuses this residual texture map with the initial reconstructed image to generate a cloud-free image with richer surface details. We employ more comprehensive evaluation metrics and conduct experiments on three public remote sensing cloud removal datasets. The experimental results demonstrate that the DBD-CR framework outperforms other GAN-based remote sensing image cloud removal methods and state-of-the-art diffusion model-based remote sensing image cloud removal methods. The main contributions of this article are as follows.

1) We propose a new cloud removal framework for optical remote sensing images. Compared to existing diffusion model-based methods, our framework places greater emphasis on the intricate details of surface features in the generated cloud-free images.
2) We employ the concept of a dual-branch diffusion model to calculate the residual texture map, thereby addressing the problem of missing surface feature details in the initial reconstructed image.
3) We design a new denoising autoencoder that incorporates a frequency domain enhancement interaction attention module to capture different frequency components of the image in the frequency domain, allowing for a better understanding of complex details. Additionally, we introduce an adaptive mixed attention module (AMAM) at the skip connection between downsampling and upsampling, effectively merging features and reducing interference from redundant information.
4) We adopt a more comprehensive set of evaluation metrics and conduct extensive experiments on three public cloud removal datasets. The results demonstrate the effectiveness and superiority of our method.

The rest of the article is organized as follows. Section II reviews related work on cloud removal methods in remote sensing images and diffusion models. Section III details our proposed DBD-CR framework and the denoising autoencoder. Section IV presents the experimental results and analysis on three public remote sensing cloud removal datasets, along with the selection of some parameters used in the experiments. Section V summarizes the article and gives conclusions.

II. Related Works

A. Cloud Removal in Remote Sensing Images

Cloud removal in remote sensing images is a critical technology in image processing [20], [21], [22], [23]. Due to various uncontrollable climatic factors, the images we obtain are often affected, with various landform elements obscured by clouds. This interference can hinder the extraction and analysis of information from subsequent remote sensing images, ultimately impacting the accuracy of downstream tasks [24].

In the early days, cloud removal techniques primarily relied on traditional image processing methods. Rossi et al. [25] employed interpolation methods for cloud removal in remote sensing images, assigning a corresponding probability to each pixel obscured by clouds or shadows. They utilized an exponential omnidirectional variogram model to describe the spatial continuity of the images. Shen et al. [26] utilized homomorphic filters to process visible light remote sensing images in the frequency domain for thin cloud removal and ground information recovery. This method semi-automatically determines the optimal cut-off frequency, allowing for the separate detection and treatment of obscured pixels, ensuring high-fidelity results. With the rise of wavelet transform, many researchers have begun to consider its application in cloud removal tasks for remote sensing images. Wavelet transform algorithms can provide both spatial and frequency information, offering significant advantages in signal processing and image analysis [27]. Zhu et al. [28] removed clouds from single remote sensing images using a dual tree complex wavelet transform. This algorithm decomposes the cloud-contaminated image into different frequency sub-bands and enhances ground object information with Laplacian filtering while employing a cloud coefficient weighting method to effectively remove the clouds. Du et al. [29] used cloud-free reference images to describe land cover and constructed a spatial variation mask using residual wavelet coefficients. After smoothing, they subtracted this mask from the contaminated images, with results demonstrating that the cloud calibration is both effective and robust.

In recent years, deep learning technologies have increasingly permeated various fields [30], [31], [32], [33], [34], [35], [36]. Due to their ability to automatically learn complex features in images and their adaptability [37], they have gradually been applied by some researchers to cloud removal tasks in remote sensing images. Li et al. [38] proposed a deep residual symmetrical concatenation network (RSC-Net) based on an encoder–decoder framework. This network includes residual convolutional and deconvolutional layers, utilizing Landsat-8 data from adjacent time periods to transform cloudy images into cloud-free images while merging relevant bands. Meraner et al. [39] proposed a deep residual neural network that combines SAR-optical data fusion to enhance image

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

reconstruction. They designed a novel cloud-adaptive loss function to maximize the preservation of original information. Ma et al. [40] employed a two-step convolutional neural network to extract cloud transparency information, enabling the recovery of surface details in areas affected by thin clouds. Although remote sensing image cloud removal methods based on convolutional neural networks can effectively extract image features and recover surface information obscured by clouds, these methods often rely on large-scale labeled data for training. In contrast, the cloud removal methods for remote sensing images based on GANs leverage the adversarial generation mechanism. Even in the absence of labeled data, these methods are able to better capture the underlying data distribution, producing more realistic and complex images. Sarukkai et al. [11] approached the problem of cloud removal as a conditional image synthesis challenge, proposing a trainable spatiotemporal generator network (STGAN) to effectively remove cloud cover. In addition, the SpAGAN framework [12] enhanced information recovery in cloudy regions by introducing a spatial attention mechanism to generate higher quality cloud-free images. Huang and Wu [13] proposed the cloud transformer GAN (CTGAN), which takes three cloudy images as input to generate corresponding cloud-free images. They also designed a feature extractor that aims to preserve the weight of cloud-free areas while reducing the weight of cloudy regions, passing the extracted features to a transformation module to identify the most critical representations.

B. Diffusion Models

Although GANs currently dominate the field of image generation [15], [41], [42], [43], the recent emergence of diffusion models has brought new ideas to the field of image generation. The DDPM [16], a seminal work on diffusion modeling, centers on the idea of generating images through a step-by-step process of noise addition and denoising. In the forward diffusion process, the model gradually adds noise to the original image to form a series of blurred images, eventually reaching complete random noise. This process can be viewed as a Markov chain in which each frame of the image depends on the state of the previous frame. The gradual addition of noise allows the model to learn the underlying distribution of the data. In the reverse denoising process, DDPM predicts how to recover the original image from the noise by training a neural network. The model denoises progressively, starting with random noise and going through several iterations to finally produce a clear image. Due to the relatively simple framework and optimization process of DDPM, it exhibits higher training stability, avoiding the common mode collapse issues associated with GANs. As a result, researchers have attempted to apply it to cloud removal tasks, achieving significant progress [17], [18], [19]. These studies gradually add noise to the original remote sensing images to simulate the effects of cloud cover. The models are then trained to learn how to recover clear images from this noise, utilizing neural networks for progressive denoising to restore surface information. Compared to previous cloud removal methods, diffusion models are better at preserving image details and structure, resulting in high-quality images.

However, generating high-fidelity cloud-free remote sensing images using DDPM typically requires thousands of sampling steps, posing significant challenges for tasks in the remote sensing field. Therefore, improving the sampling speed of diffusion models is urgently needed for remote sensing applications. Song et al. [44] proposed the denoising diffusion implicit model (DDIM), an efficient iterative implicit probability model that shares the same training process as the DDPM. DDIM achieves a deterministic generation process by extending to non-Markovian diffusion processes, enhancing the speed of sample generation. Inspired by the DDIM, the DBD-CR framework adopts a dual-branch diffusion implicit model to reconstruct cloud-free optical remote sensing images. This approach leverages the advantages of DDIM in terms of generation speed and sample quality, offering new insights into the field of cloud removal in remote sensing.

III. Method

A. Problem Definition

In the task of cloud removal, the dataset can be denoted as $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$, where each pair (x_i, y_i) represents a set of input and output data. x_i denotes the ith cloudy optical remote sensing image in the form of $x_i \in \mathbb{R}^{C\times H\times W}$. y_i denotes the cloud-free image corresponding to x_i in the form of $y_i \in \mathbb{R}^{C\times H\times W}$. C is the number of channels, H is the image height, and W is the image width. For each dataset, D can be divided into a training set D_{tr}, a validation set D_{vl}, and a test set D_{ts}, whose corresponding sets of labels do not intersect with each other, denoted as $L_{\text{tr}} \cap L_{\text{vl}} = \emptyset$, $L_{\text{tr}} \cap L_{\text{ts}} = \emptyset$, $L_{\text{vl}} \cap L_{\text{ts}} = \emptyset$, and $|L_{\text{total}}| = |L_{\text{tr}} \cup L_{\text{vl}} \cup L_{\text{ts}}|$. This dataset structure provides a foundation for supervised learning, enabling the model to learn how to recover cloud-free images y_i from cloudy images x_i. Through training, the model gradually acquires the features and patterns associated with cloud removal, improving the accuracy and quality of the generated cloud-free images.

B. Overall Framework

As shown in Fig. 1, we propose a DBD-CR, which is capable of generating high-quality cloud-free images through multiple rounds of iterative denoising and discrepancy correction. The DBD-CR framework is mainly composed of two parts. First, the initial reconstruction stage is carried out by inputting the cloudy image x_i into the denoising diffusion model, which can effectively remove the cloud occlusion by learning the mapping relation from the cloudy image to the cloud-free image to obtain the initial reconstructed image $y_i' \in \mathbb{R}^{C\times H\times W}$. Although the initial reconstructed image y_i' largely restores the original scene, the random noise used as a starting point may not be able to completely capture the fine textures and other surface features in the original cloudy image x_i, resulting in the generated cloud-free image lacking the necessary detail information and realism. Therefore, the DBD-CR framework introduces the discrepancy correction stage, where we operate the original cloudy image x_i with the initial reconstructed image y_i' as in the following equation:

$$R_i = x_i - y_i' \tag{1}$$

where R_i is the image after residual computation. This residual image R_i is input into the denoising diffusion model to obtain a residual texture map R_i', which mainly contains

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

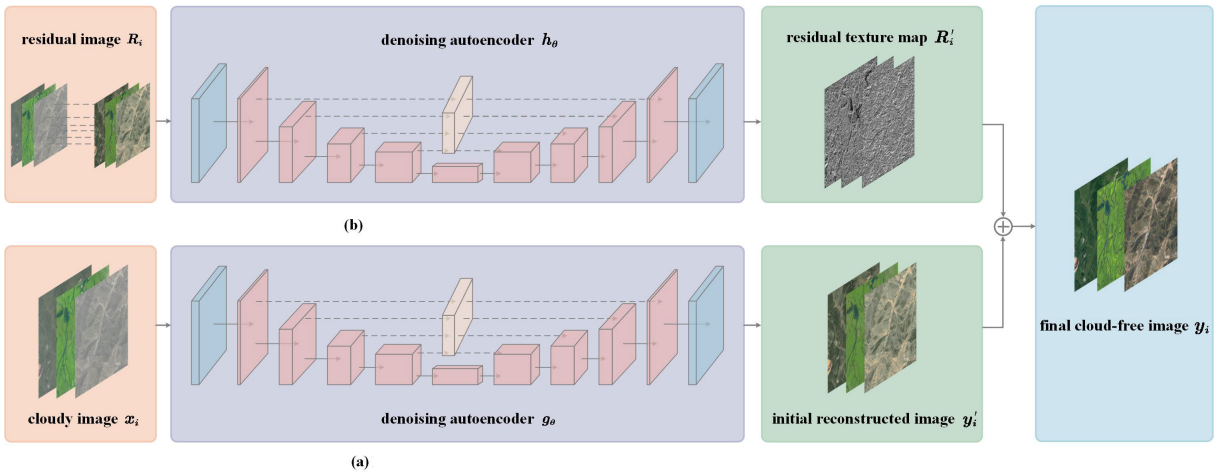


Fig. 1. Overall structure of DBD-CR. (a) Initial reconstruction stage used to generate the initial reconstructed image. (b) Discrepancy correction stage used to generate the residual texture map. g_θ and h_θ have the same overall architecture but different base channels.

some detailed information about the unreconstructed surface features. Finally, this residual texture map R_i' is fused with the initial reconstructed image y_i' as shown in the following equation to obtain the final reconstructed remote sensing cloud-free image y_i:

$$y_i = R_i' + y_i'. \tag{2}$$

C. Denoising Diffusion Model

The denoising diffusion model shows significant potential for application in remote sensing image cloud removal tasks. It outperforms traditional cloud removal techniques in terms of robustness and generative effect when dealing with complex clouds and variable surface features. In this article, the denoising diffusion model is mainly divided into two stages, the forward diffusion process and the reverse denoising process.

1) Forward Diffusion Process: The forward diffusion process gradually adds noise to the data across multiple time steps, transforming the original data distribution into a simple Gaussian noise distribution. It helps the model learn how to recover the original data from the noise. Suppose we are given a cloud-free image y_0, the forward continuous noise addition process can be regarded as a Markov chain and T denotes the maximum time step of the diffusion process. As T tends to infinity, y_T will tend to be a Gaussian distribution of random noise. Therefore, by gradually adding noise at each time step $t \in [1, T]$, we can generate the noisy image y_t as shown in the following equation:

$$y_t = \sqrt{1-\beta_t}\, y_{t-1} + \sqrt{\beta_t}\,\epsilon_t \tag{3}$$

where β_t denotes the noise variance, which typically decreases with time. ϵ_t is the noise sampled from a standard normal distribution. In this process, the image y_t of the current time step t only relies on the image y_{t-1} of the previous time step and the currently added noise ϵ_t. The specific state transfer probability can be represented by a Gaussian distribution as shown in the following equation:

$$q(y_t|y_{t-1}) = \mathcal{N}\left(y_t;\ \sqrt{1-\beta_t}\, y_{t-1},\ \beta_t \mathrm{I}\right). \tag{4}$$

Next, the relationship between y_t and the original cloud-free image y_0 at any time step t can be further derived by the superposition of normal distribution and mathematical induction, as shown in the following equation:

$$y_t = \sqrt{\bar{\alpha}_t}\, y_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon_t \tag{5}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$, and $\alpha_t = 1-\beta_t$. Similarly, we can get the distribution of y_t at any time step t, as shown in the following equation:

$$q(y_t|y_0) = \mathcal{N}\left(y_t;\ \sqrt{\bar{\alpha}_t}\, y_0,\ \left(1-\bar{\alpha}_t\right)\mathrm{I}\right). \tag{6}$$

Through the forward diffusion process, the goal is to learn how to recover the original data from the noise. This process provides the basis for the subsequent reverse denoising process, which helps the model to recover a clear image from Gaussian noise in the generation stage.

2) Reverse Denoising Process: In remote sensing images, the presence of clouds can obscure surface information. Through the reverse denoising process, it is possible to recover clear cloud-free images from cloudy ones. The traditional reverse denoising process is also a Markov chain, meaning that the data at consecutive time steps are tightly bound, preventing any leaps in prediction. However, this method has a significant drawback: it suffers from slow inference speed. Therefore, we modify the sampler to dramatically increase the sampling rate. This method breaks the Markov chain process. Assuming that $q(y_{t-1}|y_t, y_0) \sim \mathcal{N}(ky_0 + my_t, \sigma^2\mathrm{I})$ follows a normal distribution, we can express y_{t-1} as shown in the following equation:

$$y_{t-1} = ky_0 + my_t + \sigma\epsilon_t \tag{7}$$

where $\epsilon_t \sim \mathcal{N}(0, \mathrm{I})$, σ denotes the standard deviation of the noise. Next, the values of m and k are solved for by the method of coefficients to be determined using (5) in the noise addition process, and substituting them yields the new distribution expression as follows:

$$q(y_{t-1}|y_t, y_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_{t-1}}\, y_0 + \sqrt{1-\bar{\alpha}_{t-1}-\sigma^2}\,\frac{y_t - \sqrt{\bar{\alpha}_t}\, y_0}{\sqrt{1-\bar{\alpha}_t}},\ \sigma^2\mathrm{I}\right). \tag{8}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1 Training

repeat
  $y_0 \sim q(y_0)$
  $t \sim \text{Uniform}(\{1, \ldots, T\})$
  $\epsilon_t \sim \mathcal{N}(0, \text{I})$
  Take gradient descent step on
    $\nabla_\theta \| \epsilon_t - g_\theta(\sqrt{\bar{\alpha}_t} y_0 + \sqrt{1-\bar{\alpha}_t}\epsilon_t, t) \|_2^2$
until converged

Algorithm 2 Sampling

$y_T \sim \mathcal{N}(0, \text{I})$
for (t, t_p) in zip$(time, timep)$ **do**
  $y_p = \sqrt{\bar{\alpha}_p}\left(\frac{y_t - \sqrt{1-\bar{\alpha}_t} g_\theta(y_t, t)}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_p} g_\theta(y_t, t))$
end for
return y_0

Since the expression for y_0 can be introduced by (5), and ϵ_t uses the normal distribution predicted by the model, which can be introduced by (5), we can further obtain the expression for y_{t-1} as shown in the following equation:

$$y_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\left(\frac{y_t - \sqrt{1-\bar{\alpha}_t}\epsilon_t}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_{t-1}-\sigma^2}\epsilon_t + \sigma^2\epsilon \tag{9}$$

where ϵ is used for random sampling. Because the whole mathematical derivation process does not use Markov process, (3) can be considered as not needing to be strictly counted from y_t to y_{t-1}, so that y_p replaces y_{t-1}, then the formula is deformed as shown in the following equation:

$$y_p = \sqrt{\bar{\alpha}_p}\left(\frac{y_t - \sqrt{1-\bar{\alpha}_t}\epsilon_t}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_p-\sigma^2}\epsilon_t + \sigma^2\epsilon \tag{10}$$

where y_t and y_p can be separated by multiple iteration steps, $p < t-1$, which in turn achieves the effect of accelerated sampling. According to the setting in [44], the model generation is best when $\sigma = 0$. Since the following equation no longer contains any random variable ϵ, the model generation process does not introduce noise, thus making the generated image or data completely dependent on the input:

$$y_p = \sqrt{\bar{\alpha}_p}\left(\frac{y_t - \sqrt{1-\bar{\alpha}_t}\epsilon_t}{\sqrt{\bar{\alpha}_t}} + \sqrt{1-\bar{\alpha}_p}\epsilon_t\right). \tag{11}$$

As shown in Algorithms 1 and 2, they represent the whole process of denoising diffusion training and sampling. y_0 denotes the cloud-free image, and g_θ denotes the denoising model. The training stage involves calculating the model's prediction for a given case of noisy data and minimizing the gap between the predicted outputs and the real cloud-free image through continuous training until the model parameter θ converges to the completion of the training. In the sampling stage, we select a Gaussian noise sample y_T. We can sample by skipping time steps and pairing two time series together to achieve accelerated sampling through iteration.

3) Denoising Autoencoder: Denoising autoencoder plays a central role in diffusion models, directly influencing the quality and diversity of generated samples. As shown in Fig. 2(a), our denoising autoencoder is a variant built on the U-Net framework [45]. First, the image is processed through downsampling layers to extract features progressively, with each downsampling layer consisting of a frequency domain enhancement interaction attention module (FDEIAM). Next, the intermediate blocks obtained from the downsampling layers undergo symmetric upsampling operations, where the features from each downsampling layer are fused with the corresponding upsampling features. The specific fusion method involves inputting the features from each downsampling layer into an AMAM. Through skip connections, this approach allows for the direct utilization of efficient features from the downsampling stage during the upsampling stage. This characteristic helps to supplement the spatial information lost during the upsampling process. It is worth noting that while we use the same denoising autoencoder structure in the dual-branch diffusion model, the base channels differ.

a) Frequency domain enhancement interaction attention module: As shown in Fig. 2(b), we design a frequency domain enhancement interaction attention module (FDEIAM) as the basic unit in the denoising autoencoder. We take the features after passing the 1×1 convolution as the input features of FDEIAM, denoted as $F_i \in \mathbb{R}^{C\times H\times W}$, and the output features through FDEIAM are denoted as $F_o \in \mathbb{R}^{C\times H\times W}$.

First, we pass the input features through LN layers [46], 1×1 convolution, 3×3 convolution, and the ReLU activation function. Inspired by the structural design in [47], [48], and [49], we introduce a frequency domain interaction in the channel attention mechanism. In cloud removal tasks for remote sensing, the 2-D Fourier transform enables the conversion of images from the spatial domain to the frequency domain. This transformation allows for the separation of high-frequency and low-frequency information, often facilitating the processing of details that are difficult to distinguish in the spatial domain. High-frequency components typically represent the texture and details of an image, while low-frequency components correspond to flat, smooth areas [50]. Improving the high-frequency information in an image is an effective approach for reconstructing remote sensing images. Therefore, we input the features after passing the activation function into the frequency domain channel attention (FDCA). FDCA is divided into two main branches; the first branch performs 1×1 convolution after global maximum pooling of the features, and the second branch replaces the global average pooling with frequency domain interaction. Existing channel attention mechanisms [47], [48], [51], [52] focus on designing efficient methods to reduce computational overhead, often employing global average pooling for straightforward calculations. However, the simplicity of global average pooling makes it challenging for the model to capture the complex information of various inputs [49]. Therefore, we present the formula for the 2-D discrete cosine transform (DCT)

$$f_{h,w}^{2d} = \sum_{i=0}^{H-1}\sum_{j=0}^{W-1} x_{i,j}^{2d} \cos\left(\frac{\pi h}{H}\left(i+\frac{1}{2}\right)\right)\cos\left(\frac{\pi w}{W}\left(j+\frac{1}{2}\right)\right) \tag{12}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

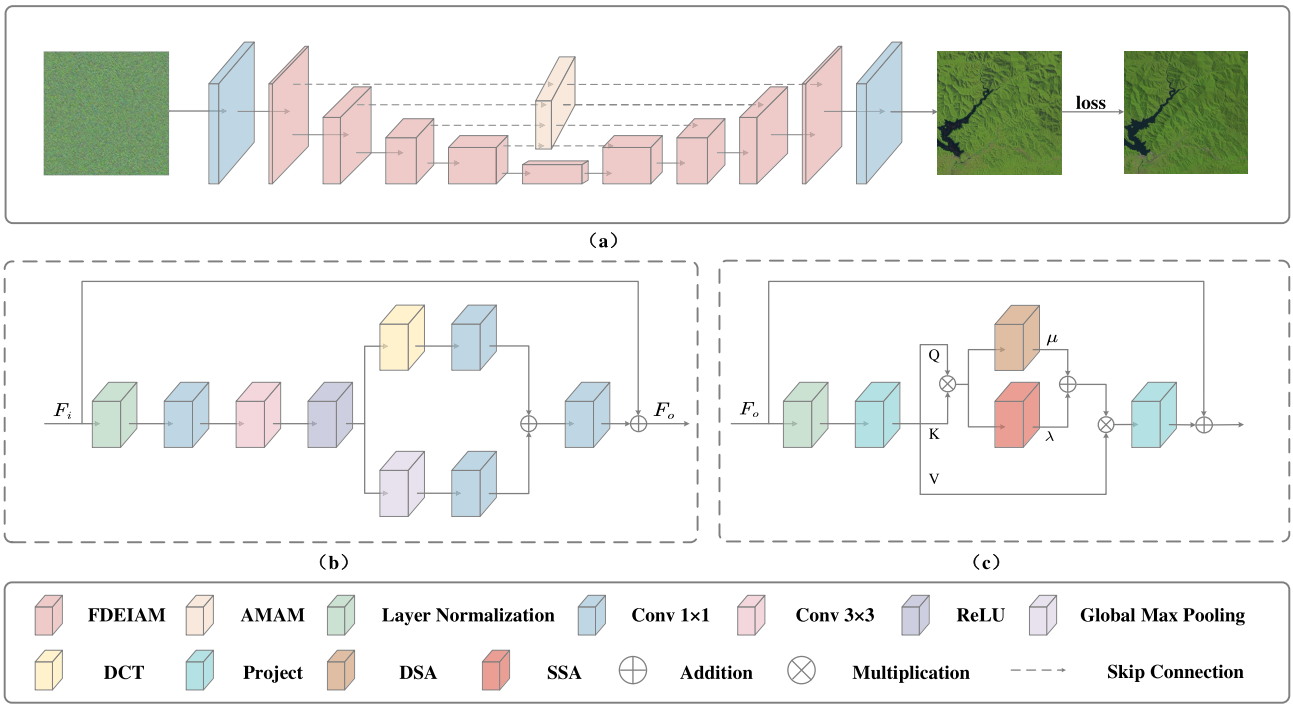


Fig. 2. (a) Overall process of generating cloud-free images from Gaussian noise maps using the denoising autoencoder architecture. (b) Specific structure of the frequency domain enhancement interactive attention module (FDEIAM). (c) Specific structure of the AMAM. The gray box at the bottom indicates annotations for the different colored blocks.

where $x^{2d} \in \mathbb{R}^{H\times W}$ is the input features, H and W are the height and width, h and w are the corresponding frequency components, and f^{2d} is the corresponding spectrum.

In the frequency domain space, the global average pooling corresponds to the lowest frequency domain component of the DCT, while the other frequency domain components are not utilized, resulting in a large amount of available information being lost. Therefore, the input feature F_a is divided into n equal parts along the channel to obtain $[F_a^0, F_a^1, \ldots, F_a^{n-1}]$, $F_a^i \in \mathbb{R}^{C_1\times H\times W}$, $i \in \{0, 1, \ldots, n-1\}$, $C_1 = (C/n)$. For each part, the 2-D DCT transform is applied and denoted as F_q^i. The results for each frequency component in the channel attention are calculated, and the top k performing frequency components are selected to be multiplied with the corresponding groups. The choice of k will be discussed in Section IV-D. Next, we apply the inverse 2-D DCT transformation on the frequency domain components as described in the following equation, followed by a 1×1 convolution operation:

$$x_{i,j}^{2d} = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1} f_{h,w}^{2d} \cos\left(\frac{\pi h}{H}\left(i+\frac{1}{2}\right)\right)\cos\left(\frac{\pi w}{W}\left(j+\frac{1}{2}\right)\right). \tag{13}$$

Finally, the features from the two branches are combined through an addition operation and then output after a 1×1 convolution.

b) Adaptive mixed attention module: As shown in Fig. 2(c), inspired by self-attention mechanisms [53], we introduce an AMAM at the skip connections of the denoising autoencoder. First, we apply layer normalization to F_o to obtain a normalized feature map $F \in \mathbb{R}^{C\times H\times W}$. Then, we generate the keys K, the queries Q, and the values V using a linear layer

$$K = FW_K, \quad Q = FW_Q, \quad V = FW_V \tag{14}$$

where W_K, W_Q, and W_V are learnable weight matrices. Then, the computation of attention can be defined as follows:

$$A = f\left(\frac{QK^T}{\sqrt{d}} + B\right)V \tag{15}$$

where d represents the dimension of the queries and keys, B is the learnable relative position bias, and $f(\cdot)$ is the scoring function used by the attention mechanism. Next, we introduce both sparse and dense attention mechanisms [54]. Sparse attention helps the model automatically select the most important features while filtering out redundant or irrelevant information, enhancing the model's performance. Dense attention ensures that the relationships between important features are fully captured, which helps the model maintain strong performance on complex tasks. The following equations present the expressions for sparse and dense attention:

$$\text{SSA} = \text{ReLU}^2\left(\frac{QK^T}{\sqrt{d}} + B\right) \tag{16}$$

$$\text{DSA} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}} + B\right). \tag{17}$$

However, while using the square of the ReLU function enhances the representation of positive features, helps to highlight important features, and reduces the effect of noise, an overly sparse attentional mechanism can lead to information loss. Therefore, we use an AMAM as shown in the following equation:

$$A_m = \left(\lambda\text{SSA} + \mu\text{DSA}\right)V \tag{18}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

where λ and μ are weighting coefficients, and the choice of specific coefficients is discussed in detail in Section IV-E.

D. Objective Function

Existing diffusion model methods [16], [44], [55] train neural networks by regressing noise predictions, enabling effective recovery of data from noise during the generation process. We define the objective function $\mathcal{L}_1$ for inputting cloudy images into the denoising diffusion model to generate the initial output images, as expressed in the following equation:

$$\mathcal{L}_1 = \mathbb{E}_{y_0 \epsilon q(y_0), \epsilon_t \sim \mathcal{N}(0,\mathrm{I})} \left\| \epsilon_t - g_\theta \left(\sqrt{\bar{\alpha}_t} y_0 + \sqrt{1-\bar{\alpha}_t} \epsilon_t, t \right) \right\|_2^2 \tag{19}$$

where g_θ represents the parameterized denoising model. Next, in the discrepancy correction stage, the objective function $\mathcal{L}_2$ is defined for obtaining the residual texture map by using the cloudy image and the initial reconstructed image through the denoising diffusion model, as expressed in the following equation:

$$\mathcal{L}_2 = \mathbb{E}_{y_0 \epsilon q(y_0), \epsilon_t \sim \mathcal{N}(0,\mathrm{I})} \left\| \epsilon_t - h_\theta \left(\sqrt{\bar{\alpha}_t} y_0 + \sqrt{1-\bar{\alpha}_t} \epsilon_t, t \right) \right\|_2^2 \tag{20}$$

where h_θ represents the parameterized denoising model. We fuse the generated residual texture map with the initial cloud-free reconstruction to obtain the final generated cloud-free remote sensing image. Therefore, our final objective function is expressed in the following equation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_1 + \mathcal{L}_2. \tag{21}$$

IV. Experiments

A. Dataset Description

We conduct experiments on three public optical remote sensing cloud removal datasets, mainly including RICE1 dataset [56], RICE2 dataset [56], and Sen2_MTC dataset [13]. In order to perform effective experimental comparisons, we follow the division of training, validation, and test sets for other comparison experiments.

The RICE1 dataset is collected from Google Earth, with a ground resolution of 5 m/pixel, and is designed to support cloud detection and removal research. It contains 500 pairs of images, each consisting of a cloudy image and its corresponding cloud-free image. All images are cropped to a size of 512 × 512 pixels, ensuring no overlap. We randomly allocate 400 pairs of images for the training set and 100 pairs for the testing set in our experiments [19].

The RICE2 dataset is derived from the Landsat 8 OLI/TIRS dataset, with a ground resolution of 30 m/pixel. It contains 736 pairs of images, each consisting of a cloudy image, a cloud-free image, and a cloud mask image, all sized at 512 × 512 pixels. We randomly allocate 588 pairs of images for the training set and 148 pairs for the testing set [19].

The Sen2_MTC dataset is collected from Sentinel-2 images and is commonly used for multitemporal training. This dataset contains 50 nonoverlapping tiles, with each tile consisting of 70 images, each sized at 256 × 256 pixels. We randomly split the dataset into training, validation, and testing sets in a ratio of 7:1:2 [13].

B. Experimental Setting

In this section, we detail the parameter settings used in the experiments and the metrics used to evaluate the final experimental results. We implement our model using the Pytorch development framework, and all experiments are performed on a single NVIDIA GeForce RTX 4090 GPU, 32 GB of RAM, with CUDA version 12.2, and optimized using the Adam optimizer. During the training stage, the batch size is set to 4, the learning rate is set to 1×10^{-4}, and the maximum noise step in the diffusion process is set to 1000, with sampling occurring every 100 time steps. For the denoising autoencoder g_θ, the base channel is set to 64, while for h_θ, the base channel is set to 32.

We employ more authoritative evaluation metrics for the generative task of cloud removal. Traditional methods based on GANs [11], [12], [13] typically use distortion evaluation metrics, which are designed to measure the loss or distortion of information during signal or image processing and usually require comparison with ground truth. As diffusion models have become a hot topic in the past couple of years, new categories of evaluation metrics have emerged. Many methods [16], [57], [58] now utilize perception evaluation metrics when assessing the quality of generated images. These metrics are designed to evaluate the visual quality of images or signals and are typically more aligned with human perceptual experiences. They not only consider technical errors or distortions but also focus on the subjective experiences of human observers during actual viewing. However, there is an inverse relationship between two types of metrics [59]; if one type is pursued excessively, the other type may suffer significantly. Moreover, it is possible for a highly realistic image completely unrelated to the input to achieve near-perfect scores on no-reference metrics. Therefore, this article utilizes both types of evaluation metrics to assess the effectiveness of DBD-CR framework. For the perception metrics, we select two of the most commonly used indicators. The first is the learned perceptual image patch similarity (LPIPS) [60], where a lower LPIPS value indicates greater similarity between two images. The second is the Fréchet inception distance (FID) [61], where a smaller FID value signifies that the images generated by the model are closer to the real data distribution. For the distortion metrics, we also select two of the most commonly used indicators. The first is the peak signal-to-noise ratio (PSNR), which measures the quality of the reconstructed image compared to the original image. A larger value of PSNR indicates better quality of the generated image. The second metric is the structural similarity index (SSIM), which measures the perceptual differences between two images by considering luminance, contrast and structural information. A larger value of SSIM indicates a better quality of the generated image.

C. Overall Experimental Results and Comparisons

In this article, we select some classical and frontier methods for cloud removal in optical remote sensing images. There are GAN-based remote sensing cloud removal methods: Pix2Pix [9], STNet [10], STGAN [11], SpAGAN [12], and CTGAN [13], as well as diffusion model-based methods DDPM-CR [17], DiffCR [18], and IDF-CR [19]. All our

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON OF EXPERIMENTAL RESULTS ON THE RICE1 DATASET

Methods	PSNR↑	SSIM↑	LPIPS↓	FID↓	Params(M)↓	MACs(G)↓
Pix2Pix[9]	26.412	0.724	0.236	83.116	11.41	58.94
STNet[10]	25.269	0.697	0.243	87.693	**4.64**	304.31
STGAN[11]	29.134	0.899	0.171	70.274	231.93	1094.94
SpAGAN[12]	29.386	0.909	0.137	57.039	185.58	389.62
CTGAN[13]	29.533	0.912	0.164	57.003	642.92	632.05
DDPM-CR[17]	29.791	0.925	0.225	60.587	445.44	852.37
DiffCR[18]	30.625	0.933	0.208	52.332	22.91	45.86
IDF-CR[19]	31.072	0.945	0.082	50.679	73.27	414.73
DBD-CR(Ours)	**31.939**	**0.953**	**0.077**	**50.292**	16.76	**37.69**

TABLE II
COMPARISON OF EXPERIMENTAL RESULTS ON THE RICE2 DATASET

Methods	PSNR↑	SSIM↑	LPIPS↓	FID↓	Params(M)↓	MACs(G)↓
Pix2Pix[9]	18.662	0.529	0.586	300.769	11.41	58.94
STNet[10]	17.193	0.506	0.603	289.332	**4.64**	304.31
STGAN[11]	22.788	0.694	0.464	287.543	231.93	1094.94
SpAGAN[12]	26.219	0.734	0.417	283.811	185.58	389.62
CTGAN[13]	26.905	0.797	0.429	261.941	642.92	632.05
DDPM-CR[17]	28.431	0.835	0.358	237.614	445.44	852.37
DiffCR[18]	29.647	0.858	0.289	220.015	22.91	45.86
IDF-CR[19]	30.762	0.893	0.265	203.197	73.27	414.73
DBD-CR(Ours)	**31.664**	**0.901**	**0.234**	**185.116**	16.76	**37.69**

TABLE III
COMPARISON OF EXPERIMENTAL RESULTS ON THE Sen2_MTC DATASET

Methods	PSNR↑	SSIM↑	LPIPS↓	FID↓	Params(M)↓	MACs(G)↓
Pix2Pix[9]	11.873	0.184	0.896	286.431	11.41	58.94
STNet[10]	11.644	0.167	0.872	284.917	**4.64**	304.31
STGAN[11]	12.749	0.256	0.886	293.323	231.93	1094.94
SpAGAN[12]	13.058	0.266	0.677	252.581	185.58	389.62
CTGAN[13]	13.381	0.270	0.629	244.638	642.92	632.05
DDPM-CR[17]	13.416	0.275	0.574	211.103	445.44	852.37
DiffCR[18]	13.782	0.292	0.531	201.982	22.91	45.86
IDF-CR[19]	13.829	0.295	0.528	196.773	73.27	414.73
DBD-CR(Ours)	**13.997**	**0.337**	**0.512**	**188.629**	16.76	**37.69**

experiments take an average of 200 rounds of test results as the final measure of model effectiveness.

As shown in Table I, we conduct experiments on the RICE1 cloud removal dataset, and the results demonstrate that the DBD-CR framework achieves the best performance. In the perception metrics, ↓ indicates that lower values are preferable. The LPIPS value of the DBD-CR framework is 0.005 lower than that of the IDF-CR framework, suggesting that a LPIPS value closer to 0 indicates greater perceptual similarity between the generated images and the ground truth. Furthermore, the FID metric for DBD-CR is also 0.387 lower compared to IDF-CR. In the distortion metrics, ↑ indicates that the higher the value, the better the result. The PSNR value of the DBD-CR framework reaches 31.939 dB, which is 0.867 dB higher than the second-best method. The SSIM value also reaches 0.953. Generally, a SSIM value closer to 1 indicates a higher structural similarity between the generated images and the ground truth. Table II presents our experimental results on the RICE2 cloud removal dataset. The DBD-CR framework achieves a reduction of 18.081 in the FID metric compared to the IDF-CR framework, indicating a significant improvement in generation quality. In the distortion metrics, the PSNR value of the DBD-CR framework is 31.664, which represents an improvement of approximately 5 dB compared to advanced GAN-based cloud removal methods. The SSIM value of the DBD-CR framework also shows an increase, improving by 0.008 compared to the second-best method. As shown in Table III, we conduct experiments on the Sen2_MTC cloud removal dataset. To maintain consistency with the experimental conditions of the first two datasets, we randomly selected one cloudy image from the Sen2_MTC dataset, denoted as $N = 1$. In the perception metrics, the DBD-CR framework achieves the lowest LPIPS value and the lowest FID value. In the distortion metrics, the DBD-CR framework reaches the highest PSNR and SSIM values. Compared to traditional GAN-based cloud removal methods, the quality of the reconstructed cloud-free images has significantly improved. In addition, we use Params and MACs to measure the parameter count and efficiency of different methods. Although the DBD-CR framework ranks third among all methods in the Params metric, it achieves significantly improved generation performance compared to the top two methods. The DBD-CR framework exhibits the best results in the MACs metric.

Therefore, the proposed DBD-CR framework demonstrates outstanding performance in optical remote sensing cloud removal tasks. To comprehensively evaluate the effectiveness of the DBD-CR framework, we employ a series of comprehensive evaluation metrics, surpassing the limitations of traditional methods that typically rely on a single metric for superiority assessment. By combining perception metrics with distortion metrics, we can provide a more comprehensive reflection of the visual quality and structural similarity of the generated images. It ensures a balanced framework performance across different aspects, highlighting the advantages and applicability of the DBD-CR framework in practical applications.

D. Selection of Frequency Component Count

In this section, we specifically discuss the selection of the optimal value k for the frequency components in FDCA. We divide each image into an 8×8 2-D DCT frequency space and perform simple calculations for each frequency component. As shown in Tables IV–VI, we conduct experiments on the selection of the optimal frequency components across three public remote sensing cloud removal datasets, where we select values of k as 1, 2, 4, 8, 16, and 32. When $k = 1$, the other branch in the frequency channel attention module is effectively equivalent to using global average pooling, as the optimal frequency component is typically the lowest frequency component. The results from the tables indicate that selecting two optimal frequency components yields better performance than selecting four or eight components. We believe that the framework may introduce too many frequency components, which can contain redundant information or noise, leading to a decline in performance. When $k = 16$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF EXPERIMENTAL RESULTS ON FREQUENCY COMPONENT COUNTS ON THE RICE1 DATASET

k	PSNR↑	SSIM↑	LPIPS↓	FID↓
1	29.422	0.806	0.195	53.619
2	31.398	0.929	0.139	51.003
4	30.827	0.901	0.141	51.476
8	30.143	0.869	0.166	51.805
16	**31.939**	**0.953**	**0.077**	**50.292**
32	31.623	0.943	0.104	50.872

TABLE V
COMPARISON OF EXPERIMENTAL RESULTS ON FREQUENCY COMPONENT COUNTS ON THE RICE2 DATASET

k	PSNR↑	SSIM↑	LPIPS↓	FID↓
1	28.197	0.707	0.411	203.673
2	30.913	0.829	0.293	193.734
4	30.721	0.802	0.378	196.242
8	30.496	0.783	0.369	198.379
16	**31.664**	**0.901**	**0.234**	**185.116**
32	31.298	0.887	0.265	188.366

TABLE VI
COMPARISON OF EXPERIMENTAL RESULTS ON FREQUENCY COMPONENT COUNTS ON THE Sen2_MTC DATASET

k	PSNR↑	SSIM↑	LPIPS↓	FID↓
1	11.332	0.224	0.683	204.319
2	13.277	0.306	0.561	194.831
4	13.056	0.299	0.574	196.443
8	12.741	0.273	0.606	198.236
16	**13.997**	**0.337**	**0.512**	**188.629**
32	13.624	0.325	0.537	191.245

the DBD-CR framework generally achieves the lowest perception metrics and the highest distortion metrics. This selection effectively identifies the most efficient frequency combination, capturing the textures of certain surface features. However, as k continues to increase to 32, the quality of the generated images declines.

Therefore, the optimal number of frequency components k selected in this article is 16. We find that relying solely on the lowest frequency components is insufficient to fully represent the content of the image. While low-frequency information is crucial for image restoration tasks, it fails to capture high-frequency details, such as surface texture features. In addition, as the number of optimal frequency components increases, it can lead to information redundancy between the frequency channels, which also increases computational complexity, reducing the performance of the DBD-CR framework.

E. Selection of AMAM Weighting Factors

In this section, we discuss the sparse attention weight coefficient λ and the dense attention weight coefficient μ in the AMAM to obtain the optimal configuration. We select

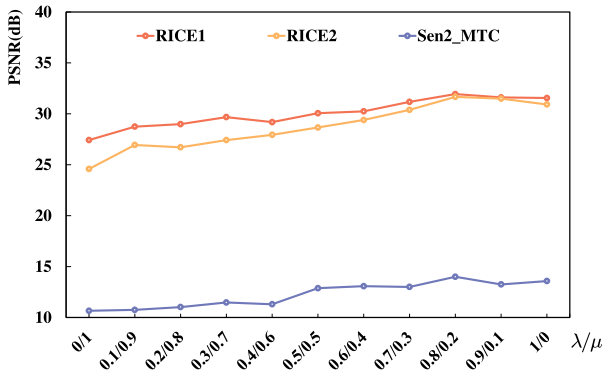


Fig. 3. Experimental results of PSNR metrics with different weight coefficients on three datasets.

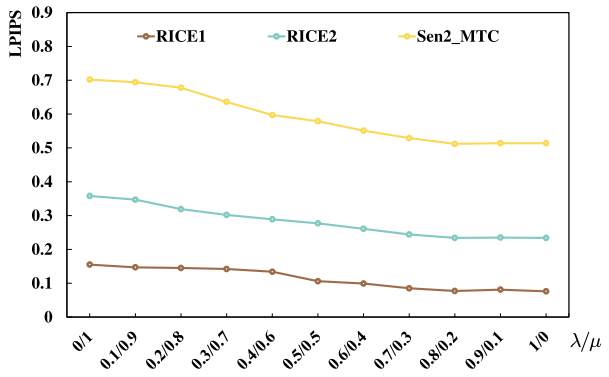


Fig. 4. Experimental results of LPIPS metrics with different weight coefficients on three datasets.

the PSNR metric from the distortion metrics and the LPIPS metric from the perception metrics as the final criteria for selecting the weight coefficients. We conduct experiments on three public remote sensing cloud removal datasets. As shown in Fig. 3, the horizontal axis represents the ratio of the sparse attention weight λ to the dense attention weight μ, while the vertical axis shows the PSNR values. We select 11 different scenarios, ensuring that $\lambda + \mu = 1$. The results from the figure indicate that as λ increases by 0.1 and μ decreases by 0.1, the PSNR values also increase. Although there is a brief decline when $\lambda = 0.4$ and $\mu = 0.6$, the overall trend remains upward until $\lambda = 0.8$ and $\mu = 0.2$. Subsequently, as λ continues to increase by 0.1 and μ decreases by 0.1, the PSNR values begin to decrease. As shown in Fig. 4, the horizontal axis represents the ratio of the sparse attention weight λ to the dense attention weight μ, while the vertical axis shows the LPIPS values. We also select 11 different scenarios, ensuring that $\lambda + \mu = 1$. The experimental results from the figure show that as λ increases by 0.1 and μ decreases by 0.1, the LPIPS values consistently decrease until $\lambda = 0.8$ and $\mu = 0.2$. Subsequently, as λ continues to increase by 0.1 and μ continues to decrease by 0.1, the line plot stabilizes.

Therefore, based on the experimental results from both types of metrics, we select 0.8 as the sparse attention weight coefficient and 0.2 as the dense attention weight coefficient in the AMAM. We believe that excessive reliance on the sparse attention mechanism may cause important detail information to be overlooked, while excessive use of the dense attention

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

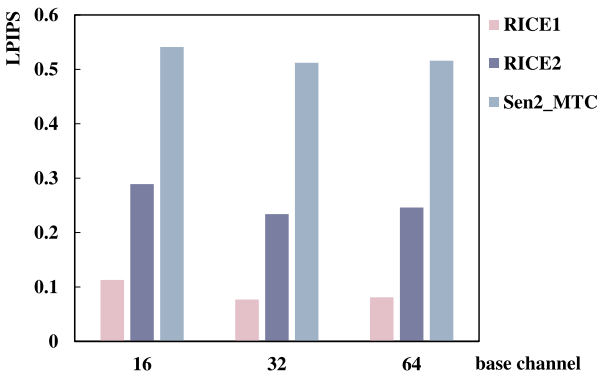


Fig. 5. Experimental results of LPIPS metrics with different h_θ base channels on three datasets.

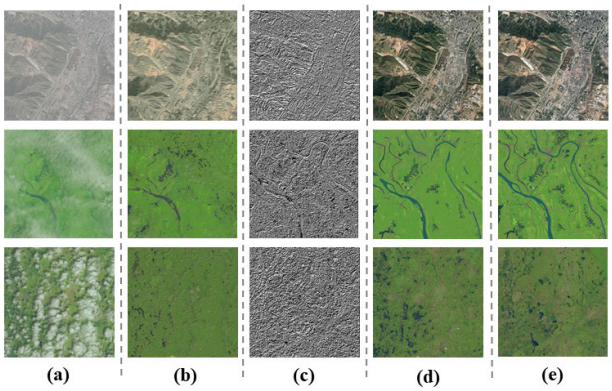


Fig. 6. Visualization comparison between the initial reconstruction stage and the discrepancy correction stage. The figure randomly selects three cloudy images with different cloud distributions for validation. (a) Three randomly selected cloudy images with different cloud distributions. (b) Corresponding initial reconstructed images. (c) Corresponding residual texture images. (d) Final generated cloud-free images. (e) Cloud-free image labels.

mechanism can lead the framework to indiscriminately focus on all features, introducing redundant information. Therefore, by combining the two attention mechanisms with specific weight coefficients, we can better integrate the downsampled features with the upsampled features, achieving optimal performance for the DBD-CR framework.

F. Comparison Experiments of h_θ Base Channel

In this section, we specifically discuss the impact of the discrepancy correction stage on the DBD-CR framework. As shown in Fig. 5, we conduct comparative experiments on the performance of the denoising autoencoder h_θ with different base channel counts. We select base channel counts of 16, 32, and 64 for h_θ and conduct experiments on three datasets, using LPIPS as the evaluation metric. The results show that when the base channel count of h_θ is 16, the LPIPS value is significantly higher than in the other two cases. It indicates that when the base channel count is too low, the framework's expressive capability is relatively weak, affecting the reconstruction quality of the residual textures. As the base channel count increases, h_θ can capture important texture features better. However, an increase in the base channel count may also lead to a rise in the number of parameters, potentially resulting in overfitting. When the base channel count is 32, the LPIPS value is not significantly different from that of the 64-channel case, indicating that the model achieves a good balance between texture reconstruction and generalization ability at this point. As shown in Fig. 6, regardless of the cloud distribution type, the initial reconstructed images are blurred due to cloud interference, while the final generated cloud-free images showcase richer details and clearer textures.

Therefore, we choose to set the base channel count of h_θ to 32. It further indicates that reconstructing the residual textures is not only easier but also a key step in generating high-quality cloud-free images within the DBD-CR framework.

G. Visualization Comparison of Cloud Removal Effects

To visually demonstrate the quality of the cloud-free images generated by the DBD-CR framework, we conduct a visualization comparison experiment for cloud removal effects. As shown in Fig. 7, the first row displays, from left to right, a randomly selected cloudy image, the ground truth, the initial reconstructed image, and the image generated by the DBD-CR framework. The second and third rows present the eight methods we compared. Through comparative experiments, we find that the DBD-CR framework demonstrates higher detail complexity and realism in image generation, particularly excelling in capturing complex details of topographical features compared to other methods. The images generated by the DBD-CR framework show high similarity in color to the ground truth, while GAN-based cloud removal methods in remote sensing typically restore topographical features using simple filling patterns, leading to a lack of coherence in some generated images.

TABLE VII
ABLATION STUDY RESULTS ON THE RICE1 DATASET

Model			PSNR↑	SSIM↑	LPIPS↓	FID↓
FDEIAM	AMAM	Residual				
×	×	×	27.738	0.887	0.225	63.449
✓	×	×	30.335	0.922	0.186	55.364
✓	✓	×	31.127	0.942	0.109	52.043
✓	✓	✓	**31.939**	**0.953**	**0.077**	**50.292**

TABLE VIII
ABLATION STUDY RESULTS ON THE RICE2 DATASET

Model			PSNR↑	SSIM↑	LPIPS↓	FID↓
FDEIAM	AMAM	Residual				
×	×	×	26.815	0.819	0.331	216.177
✓	×	×	28.226	0.833	0.273	202.701
✓	✓	×	30.453	0.866	0.252	193.587
✓	✓	✓	**31.664**	**0.901**	**0.234**	**185.116**

H. Ablation Experiments

In this section, to further demonstrate the impact of the frequency domain enhancement interactive attention module (FDEIAM), the AMAM, and the residual textures from the discrepancy correction stage (Residual) on the performance of the DBD-CR framework, we conduct ablation experiments. We use four metrics to evaluate the results, as shown in Tables VII–IX. On the three remote sensing cloud removal datasets, incorporating the FDEIAM leads to an improvement

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

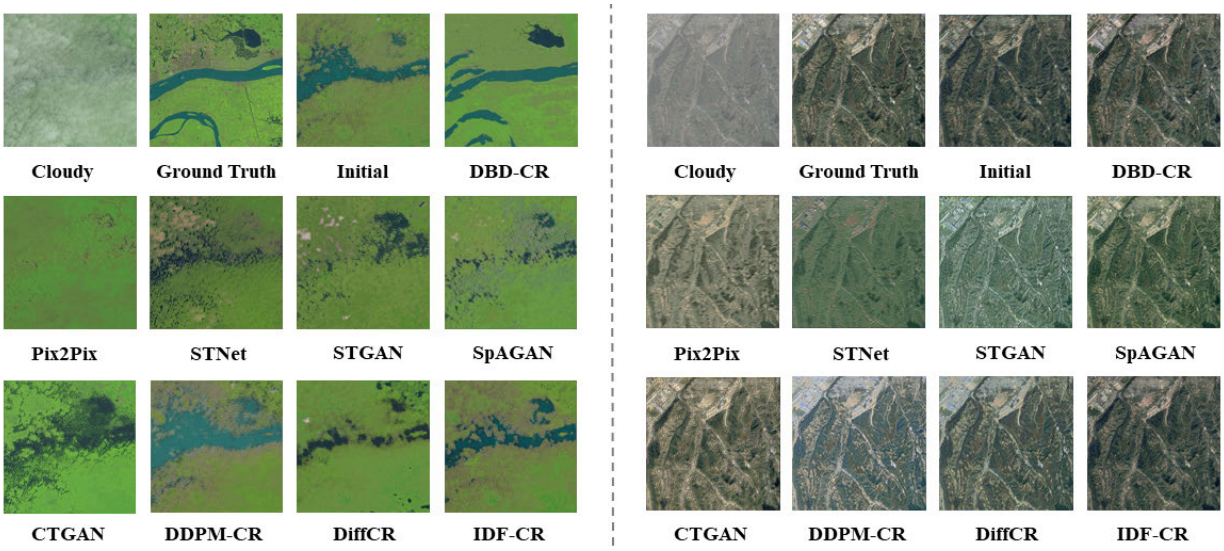


Fig. 7. Visualization of experimental results comparing the DBD-CR framework with other methods. The figure randomly selects two different terrain categories from two datasets.

TABLE IX
ABLATION STUDY RESULTS ON THE Sen2_MTC DATASET

Model			PSNR↑	SSIM↑	LPIPS↓	FID↓
FDEIAM	AMAM	Residual				
×	×	×	12.998	0.273	0.596	219.104
✓	×	×	13.379	0.294	0.573	209.877
✓	✓	×	13.603	0.311	0.544	193.625
✓	✓	✓	**13.997**	**0.337**	**0.512**	**188.629**

in the quality of the generated images compared to using only the single-branch diffusion model, with an increase of approximately 2 dB in PSNR and about 0.3 in SSIM. Next, after incorporating the AMAM, the perception metrics generally decrease, with the LPIPS metric reducing by approximately 0.05. The decrease in FID is particularly noticeable, especially on the RICE2 and Sen2_MTC datasets. Finally, by incorporating the residual textures from the discrepancy correction stage, the quality of the generated images is further enhanced. The perception and distortion metrics show that the DBD-CR framework achieves optimal results.

Thus, FDEIAM selects the most important frequency components for image reconstruction by applying DCT within the channel attention mechanism, suppressing noise interference and enhancing the detail quality of the generated images. AMAM combines the advantages of sparse and dense attention, adaptively selecting and integrating relevant features, which increases the model's flexibility and adaptability. The residual texture in the difference correction stage helps the model focus on reconstructing subtle differences, improving the perceptual similarity and detail reconstruction capability of the images. By integrating these three modules, the DBD-CR framework achieves higher generation quality and stronger generalization ability in the task of cloud removal for remote sensing images.

I. Discussion of Limitations

Although the DBD-CR framework demonstrates excellent performance compared to other methods, our approach still has some limitations. As shown in Fig. 8, the results in the figure indicate that when the input cloudy image primarily consists of low-texture areas like deserts, our method encounters some challenges in reconstructing the cloud-free image. We believe that the surface features in these areas usually exhibit low-frequency dominant smooth transitions. In these regions, subtle high-frequency components, such as edges and textures, are extremely sparse, making it more challenging to extract high-frequency information. The DBD-CR framework relies on frequency domain enhancement techniques to extract these high-frequency details. However, when there is insufficient high-frequency information in the original signal, the DBD-CR framework may misinterpret noise or low-frequency interference from cloud residuals as valid signals while attempting to enhance the high-frequency components to restore details. This further leads to distorted reconstruction results that do not accurately reflect the surface features of the desert. In addition, while the non-Markovian inference design of the DBD-CR framework alleviates the computational pressure of single-step calculations, the inherent iterative dependencies during the diffusion model training stage remain an efficiency bottleneck. In the future, we will explore more efficient iterative mechanisms to achieve optimal efficiency while maintaining high accuracy.

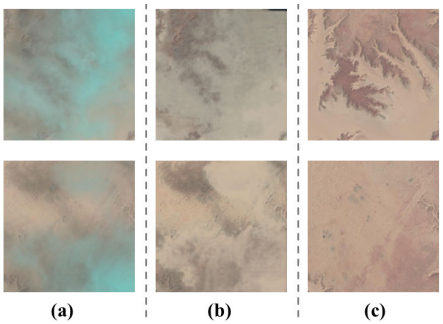


Fig. 8. Two examples of limitations. The DBD-CR framework shows deficiencies when the input cloudy image consists of low-texture areas, such as deserts. (a) Cloudy image. (b) Cloud-free image generated by the DBD-CR framework. (c) Corresponding ground truth.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this article, we propose a dual-branch diffusion residual reconstruction framework for cloud removal in optical remote sensing images (DBD-CR framework). Compared to traditional GAN-based cloud removal methods and advanced diffusion model-based methods, the DBD-CR framework places greater emphasis on reconstructing the complex details of surface features in cloud-free images to generate high-fidelity images for subsequent downstream tasks. It designs a dual-branch diffusion model architecture where one branch is used to generate the initial reconstructed cloud-free image, and the other branch is used to compute the residual texture maps, which are fused to obtain the rich detail cloud-free image. Finally, the experimental results demonstrate the effectiveness and superiority of the DBD-CR framework through comprehensive validation on three public remote sensing datasets.

However, obtaining real cloudy and cloud-free data pairs in the real world remains a significant challenge. We hope to leverage unsupervised learning in the future to learn the underlying structure of data for generating high-quality cloud-free images, further advancing the field of remote sensing cloud removal.

References

[1] H. Huang et al., "Object-based attention mechanism for color calibration of UAV remote sensing images in precision agriculture," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3043980.

[2] L. Wu, X. Liu, M. Liu, J. Yang, L. Zhu, and B. Zhou, "Online forest disturbance detection at the sub-annual scale using spatial context from sparse Landsat time series," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507814.

[3] N. Longbotham, C. Chaapel, L. Bleiler, C. Padwick, W. J. Emery, and F. Pacifici, "Very high resolution multiangle urban classification analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 4, pp. 1155–1170, Apr. 2012.

[4] H. Li, B. Zhou, and F. Xu, "Variation analysis of spectral characteristics of reclamation vegetation in a rare Earth mining area under environmental stress," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408412.

[5] M. D. King, S. Platnick, W. P. Menzel, S. A. Ackerman, and P. A. Hubanks, "Spatial and temporal distribution of clouds observed by MODIS onboard the Terra and Aqua satellites," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 7, pp. 3826–3852, Jul. 2013.

[6] C. Ledig et al., "Photo-realistic single image super-resolution using a generative adversarial network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4681–4690.

[7] Z. Wan et al., "Bringing old photos back to life," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 2747–2757.

[8] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 2223–2232.

[9] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 1125–1134.

[10] Y. Chen, Q. Weng, L. Tang, X. Zhang, M. Bilal, and Q. Li, "Thick clouds removing from multitemporal Landsat images using spatiotemporal neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4400214.

[11] V. Sarukkai, A. Jain, B. Uzkent, and S. Ermon, "Cloud removal from satellite images using spatiotemporal generator networks," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, Mar. 2020, pp. 1796–1805.

[12] H. Pan, "Cloud removal for remote sensing imagery via spatial attention generative adversarial network," 2020, *arXiv:2009.13015*.

[13] G.-L. Huang and P.-Y. Wu, "CTGAN: Cloud transformer generative adversarial network," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 511–515.

[14] V. Sarukkai, A. Jain, B. Uzkent, and S. Ermon, "Cloud removal in satellite images using spatiotemporal generative networks," 2019, *arXiv:1912.06838*.

[15] K. Enomoto et al., "Filmy cloud removal on satellite imagery with multispectral conditional generative adversarial nets," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, Jul. 2017, pp. 48–56.

[16] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. NIPS*, vol. 33, 2020, pp. 6840–6851.

[17] R. Jing, F. Duan, F. Lu, M. Zhang, and W. Zhao, "Denoising diffusion probabilistic feature-based network for cloud removal in Sentinel-2 imagery," *Remote Sens.*, vol. 15, no. 9, p. 2217, Apr. 2023.

[18] X. Zou et al., "DiffCR: A fast conditional diffusion framework for cloud removal from optical satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612014.

[19] M. Wang, Y. Song, P. Wei, X. Xian, Y. Shi, and L. Lin, "IDF-CR: Iterative diffusion process for divide-and-conquer cloud removal in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5615014.

[20] C. Long, X. Li, Y. Jing, and H. Shen, "Bishift networks for thick cloud removal with multitemporal remote sensing images," *Int. J. Intell. Syst.*, vol. 2023, no. 1, Jan. 2023, Art. no. 9953198.

[21] Q. Bie and X. Su, "MCGFE-CR: Cloud removal with multiscale context-guided feature enhancement network," *IEEE Access*, vol. 12, pp. 181303–181315, 2024.

[22] T. Liu, Y. Liu, C. Zhang, L. Yuan, X. Sui, and Q. Chen, "Hyperspectral image super-resolution via dual-domain network based on hybrid convolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512518.

[23] H. Wei et al., "Spatio-temporal feature fusion and guide aggregation network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5642216.

[24] Z. C. Tan, X. F. Du, W. Man, X. Z. Xie, G. S. Wang, and Q. Nie, "Unsupervised remote sensing image thin cloud removal method based on contrastive learning," *IET Image Process.*, vol. 18, no. 7, pp. 1844–1861, May 2024.

[25] R. E. Rossi, J. L. Dungan, and L. R. Beck, "Kriging in the shadows: Geostatistical interpolation for remote sensing," *Remote Sens. Environ.*, vol. 49, no. 1, pp. 32–40, Jul. 1994.

[26] H. Shen, H. Li, Y. Qian, L. Zhang, and Q. Yuan, "An effective thin cloud removal procedure for visible remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 96, pp. 224–235, Oct. 2014.

[27] D. Zhang, "Wavelet transform," in *Fundamentals of Image Data Mining: Analysis, Features, Classification and Retrieval*. Cham, Switzerland: Springer, 2019, pp. 35–44, doi: 10.1007/978-3-030-17989-2_3.

[28] X. Zhu et al., "Cloud removal for single images based on dual tree complex wavelet transform," in *Proc. 10th Int. Conf. Signal Process. Commun. Syst. (ICSPCS)*, Dec. 2016, pp. 1–6.

[29] Y. Du, B. Guindon, and J. Cihlar, "Haze detection and removal in high resolution satellite image with wavelet analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 1, pp. 210–217, Jan. 2002.

[30] C. Yan et al., "Discriminative correspondence estimation for unsupervised RGB-D point cloud registration," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 2, pp. 1209–1223, Feb. 2025.

[31] Y. Cai, X. Sui, G. Gu, and Q. Chen, "Multi-modal interaction with token division strategy for RGB-T tracking," *Pattern Recognit.*, vol. 155, Nov. 2024, Art. no. 110626.

[32] X. Song, D. Han, C. Chen, X. Shen, and H. Wu, "Vman: Visual-modified attention network for multimodal paradigms," *Vis. Comput.*, vol. 41, no. 4, pp. 2737–2754, Mar. 2025.

[33] G. Chen, H. Lu, D. Di, L. Li, M. Emam, and W. Jing, "StfMLP: Spatiotemporal fusion multilayer perceptron for remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[34] Y. Zhang, Z. Gao, X. Wang, and Q. Liu, "Image representations of numerical simulations for training neural networks," *Comput. Model. Eng. Sci.*, vol. 134, no. 2, pp. 821–833, 2023.

[35] X. Shi, Y. Zhang, A. Pujahari, and S. K. Mishra, "When latent features meet side information: A preference relation based graph neural network for collaborative filtering," *Expert Syst. Appl.*, vol. 260, Jan. 2025, Art. no. 125423.

[36] D. Guan, Y. Xing, J. Huang, A. Xiao, A. El Saddik, and S. Lu, "S2Match: Self-paced sampling for data-limited semi-supervised learning," *Pattern Recognit.*, vol. 159, Mar. 2025, Art. no. 111121.

[37] Z. Wang, Z. Liao, B. Zhou, G. Yu, and W. Luo, "SwinURNet: Hybrid transformer-CNN architecture for real-time unstructured road segmentation," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.

[38] W. Li, Y. Li, D. Chen, and J. C.-W. Chan, "Thin cloud removal with residual symmetrical concatenation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 153, pp. 137–150, Jul. 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.

[39] A. Meraner, P. Ebel, X. X. Zhu, and M. Schmitt, "Cloud removal in Sentinel-2 imagery using a deep residual neural network and SAR-optical data fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 333–346, Aug. 2020.
[40] D. Ma, R. Wu, D. Xiao, and B. Sui, "Cloud removal from satellite images using a deep learning model with the cloud-matting method," *Remote Sens.*, vol. 15, no. 4, p. 904, Feb. 2023.
[41] Y. Zhao, S. Shen, J. Hu, Y. Li, and J. Pan, "Cloud removal using multimodal GAN with adversarial consistency loss," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[42] Y. Hao, W. Jiang, W. Liu, Y. Li, and B.-D. Liu, "Selecting information fusion generative adversarial network for remote-sensing image cloud removal," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[43] C. Zhang, X. Zhang, Q. Yu, and C. Ma, "An improved method for removal of thin clouds in remote sensing images by generative adversarial network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 6706–6709.
[44] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," 2020, *arXiv:2010.02502*.
[45] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, Munich, Germany. Cham, Switzerland: Springer, Oct. 2015, pp. 234–241.
[46] J. Lei Ba, J. Ryan Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.
[47] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[48] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11534–11542.
[49] Z. Qin, P. Zhang, F. Wu, and X. Li, "FcaNet: Frequency channel attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 783–792.
[50] S. Guo, H. Yong, X. Zhang, J. Ma, and L. Zhang, "Spatial-frequency attention for image denoising," 2023, *arXiv:2302.13598*.
[51] H. Lee, H.-E. Kim, and H. Nam, "SRM: A style-based recalibration module for convolutional neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1854–1862.
[52] X. Li, W. Wang, X. Hu, and J. Yang, "Selective kernel networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 510–519.
[53] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jun. 2017, pp. 5998–6008.
[54] S. Zhou, D. Chen, J. Pan, J. Shi, and J. Yang, "Adapt or perish: Adaptive sparse transformer with attentive feature refinement for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 2952–2963.
[55] C. Saharia, J. Ho, W. Chan, T. Salimans, D. J. Fleet, and M. Norouzi, "Image super-resolution via iterative refinement," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 4, pp. 4713–4726, Apr. 2022.
[56] D. Lin, G. Xu, X. Wang, Y. Wang, X. Sun, and K. Fu, "A remote sensing image dataset for cloud removal," 2019, *arXiv:1901.00600*.
[57] A. Nichol et al., "GLIDE: Towards photorealistic image generation and editing with text-guided diffusion models," 2021, *arXiv:2112.10741*.
[58] C. Saharia et al., "Photorealistic text-to-image diffusion models with deep language understanding," in *Proc. NIPS*, vol. 35, 2022, pp. 36479–36494.
[59] Y. Blau and T. Michaeli, "The perception-distortion tradeoff," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6228–6237.
[60] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 586–595.
[61] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local nash equilibrium," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, Jan. 2017, pp. 1–12.



Chen Yang (Member, IEEE) received the B.S. degree from Changchun University of Science and Technology, Changchun, Jilin, China, in 2021, and the M.S. degree from Jilin University, Changchun, in 2024, where she is currently pursuing the Ph.D. degree with the College of Computer Science and Technology.

Her research interests include computer vision, remote sensing, and diffusion model.



Qinxu Gao received the B.S. degree from Qingdao University of Technology, Qingdao, Shandong, China, in 2022. He is currently pursuing the M.S. degree with the College of Computer Science and Technology, Jilin University, Changchun, Jilin, China.

His research interests include computer vision and deep learning.



Guanhua Chen received the B.S. degree from Changchun University of Science and Technology, Changchun, Jilin, China, in 2022. He is currently pursuing the M.S. degree with the College of Computer Science and Technology, Jilin University, Changchun.

His research interests include image processing and computer vision.



Huiying Li received the B.S., M.S., and Ph.D. degrees in computer science and technology from Jilin University, Changchun, China, in 2001, 2004, and 2008, respectively.

She is currently an Associate Professor with Jilin University. Her research interests include computer image processing and pattern recognition.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:59:16 UTC from IEEE Xplore. Restrictions apply.